Lesson 7.1 - Continuous Probability Distributions

> **Video:** Continuous Probability Distributions - Basic Introduction by The Organic Chemistry Tutor
> **Chapter:** 7 - Continuous Distributions

Continuous random variables

A **continuous** random variable X can take **any** value in a range, such as 2, 3.5 or 4.68. A **discrete** one is restricted to specific values, such as 1, 2, 3, ...

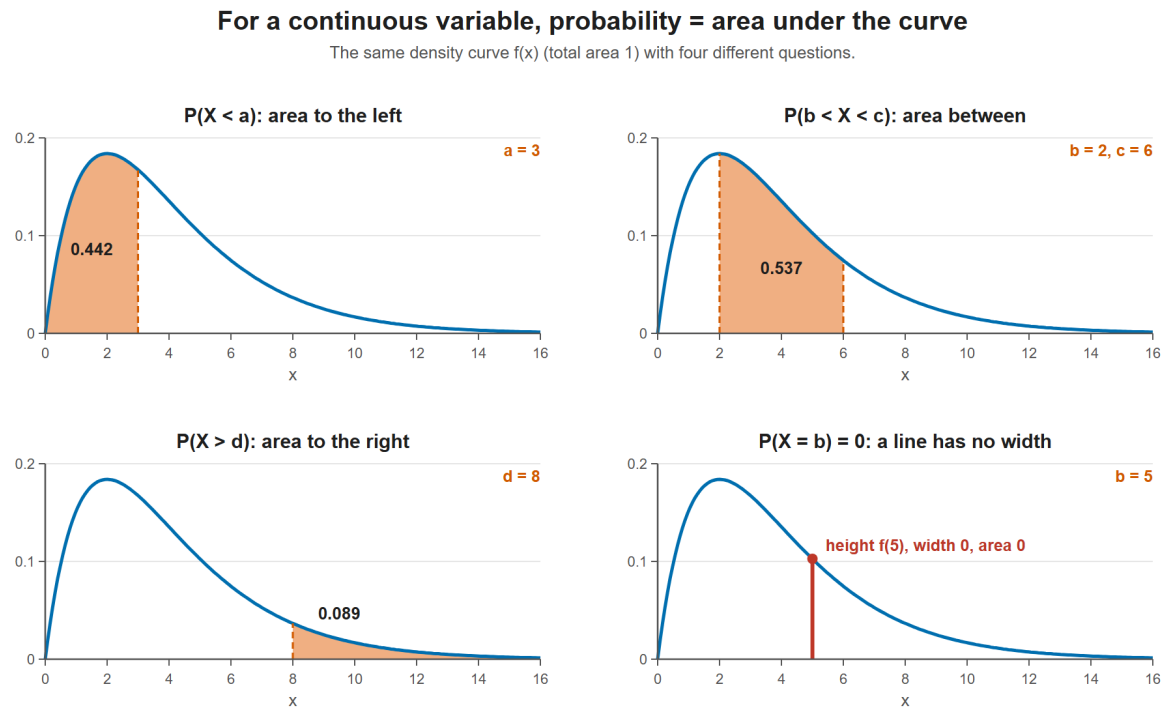
The curve of a continuous distribution is its **probability density function (PDF)**, $f(x)$. Its height above the x-axis at x is $f(x)$.

The key rules

1. $f(x) \geq 0$ everywhere: the curve never dips below the x-axis.
2. **Total area under the curve = 1.**
3. **Probability = area under the curve** over the interval you want:
   - $P(X < a)$ = area to the **left** of a
   - $P(b < X < c)$ = area **between** b and c
   - $P(X > d)$ = area to the **right** of d
4. **A single value has zero probability:** $P(X = b) = 0$. A vertical line has height but no width, so its area is 0.
5. Therefore $<$ and $\leq$ give the **same** probability:

$$P(X < a) = P(X \leq a), \qquad P(b < X < c) = P(b \leq X \leq c)$$

The **normal distribution** (bell curve) is the most important example. It is covered in Chapter 8.



Because a single value has zero area, P(X < a) = P(X ≤ a) for any continuous distribution.

Every probability question about a continuous variable is a shaded area under the density curve. A single value is a vertical line with no width, so its probability is 0.

Uniform distribution

$f(x)$ is **constant** between a and b, so the graph is a rectangle with area 1.

Example: a uniform distribution from $a = 2$ to $b = 6$ has base $= 4$, so the height must be $\frac{1}{4}$ for the area to equal $4 \times \frac{1}{4} = 1$.

PDF	Mean	Standard deviation
$f(x) = \frac{1}{b-a}$ for $a \le x \le b$	$\mu = \frac{a+b}{2}$	$\sigma = \frac{b-a}{\sqrt{12}}$

Exponential distribution

The curve starts at height λ on the y-axis and **decays** as x increases.

PDF	Rate	Left area	Right area
$f(x) = \lambda e^{-\lambda x}$	$\lambda = \frac{1}{\mu}$	$P(X < x) = 1 - e^{-\lambda x}$	$P(X > x) = e^{-\lambda x}$

Key takeaways

- Continuous distributions are described by a PDF $f(x) \ge 0$ with a total area of 1.
- **Probability = area under the curve.**

- $P(X = \text{one exact value}) = 0$, so "$<$" and "$\leq$" are interchangeable.
- Uniform: a flat rectangle at height $1/(b-a)$. Exponential: a decaying curve with $\lambda = 1/\mu$.